

SOFTWARE REQUIREMENTS SPECIFICATION

HOTEL BOOKING SYSTEM

SRS Document with UML Diagrams

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- Software Requirements Specification
 - Functional and Non-Functional Requirements
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1. Introduction

1.1 Purpose

The purpose of the **Hotel Booking System** is to provide an online platform for customers to search hotels, check room availability, book rooms, make payments, and manage their bookings. The system also helps hotel administrators manage rooms, customers, pricing, bookings, and billing efficiently.

1.2 Scope

- Customer registration and login
- Hotel and room search
- Checking room availability
- Online room booking
- Seasonal/dynamic pricing management
- Customer billing and online payment
- Booking cancellation and booking history
- Admin management of hotels, rooms, prices and customers
- Generation of booking, payment and revenue reports

1.3 Definitions and Abbreviations

Term	Description
SRS	Software Requirements Specification
Admin	Person who manages the hotel system
Customer	Person who books a hotel room
UI	User Interface
DB	Database
API	Application Programming Interface
Booking ID	Unique identification number for a booking

2. Overall Description

2.1 Product Perspective

The Hotel Booking System is a web/mobile-based application consisting of **Frontend** → **Backend/API** → **Database**. The frontend allows customers and administrators to interact with the system. The backend processes requests and business logic, while the database stores customer, room, booking, payment and pricing information.

2.2 Product Functions

- User registration and login
- Hotel/room search
- Room availability checking
- Room booking
- Price calculation
- Seasonal pricing
- Payment processing
- Invoice generation
- Booking cancellation
- Customer booking history
- Room management
- Customer management
- Report generation

2.3 User Classes

User	Main Responsibilities
Customer	Register/login, search rooms, check availability, book, pay, view/cancel bookings, view invoice
Administrator	Manage rooms, availability, seasonal prices, bookings, customers, payments and reports

2.4 Operating Environment

Frontend: HTML, CSS, JavaScript / React

Backend: Node.js and Express.js

Database: MySQL / MongoDB

Server: Web server or cloud server

Browsers: Chrome, Firefox, Edge and Safari

Operating systems: Windows, Linux, Android and iOS

3. Functional Requirements

FR1 – User Registration

The system shall allow new customers to create an account using name, email, phone number and password, with validation before account creation.

FR2 – User Login

The system shall allow registered customers and administrators to log in using valid credentials.

FR3 – Hotel/Room Search

Customers shall search using location, check-in date, check-out date, number of guests and room type.

FR4 – Room Availability

The system shall display available rooms for selected dates and prevent double booking.

FR5 – Room Details

The system shall display room number, type, capacity, facilities, price, images and availability status.

FR6 – Room Booking

Customers shall select an available room and create a booking; a unique Booking ID shall be generated.

FR7 – Seasonal Pricing

Administrators shall define normal, peak, festival and holiday pricing; the applicable rate shall be calculated from booking dates.

FR8 – Billing

The system shall calculate room charges, taxes/other charges, discounts and final amount.

FR9 – Payment

Customers shall make payments through supported payment methods and the system shall update payment status.

FR10 – Booking Confirmation

After successful booking/payment, the system shall provide Booking ID, customer, room, dates and payment status.

FR11 – Cancellation

Customers shall cancel eligible bookings according to the hotel's cancellation policy.

FR12 – Booking History

Customers shall view current, previous and cancelled bookings.

FR13 – Admin Room Management

Administrators shall add, update, remove rooms and change availability and prices.

FR14 – Customer Management

Administrators shall view and manage registered customer information.

FR15 – Reports

Administrators shall generate booking, availability, cancellation, revenue, customer and payment reports.

4. Non-Functional Requirements

4.1 Performance

The system should respond quickly, display search results within a reasonable time, and support multiple users simultaneously.

4.2 Security

Passwords must be securely stored; only authorized administrators may access admin functions; customer and payment information must be protected.

4.3 Reliability

The system should operate reliably with minimum downtime, prevent loss of booking information, and maintain database backups.

4.4 Usability

The interface should be simple and user-friendly, with clear navigation and error messages.

4.5 Availability

The system should be available for searching and booking subject to server and maintenance availability.

4.6 Maintainability

The software should be modular so developers can modify or add features easily.

4.7 Scalability

The system should support increasing numbers of customers, hotels, rooms, bookings and transactions.

5. External Interface Requirements

5.1 User Interface

Home page, Login, Registration, Search, Room Details, Booking, Payment, Booking Confirmation, Customer Dashboard and Admin Dashboard.

5.2 Hardware Interface

The system can be accessed using desktop computers, laptops, smartphones and tablets.

5.3 Software Interface

Database server, payment gateway, email/SMS notification service and web/application server.

5.4 Communication Interface

Secure internet communication such as HTTPS shall be used between client and server.

6. System and Database Requirements

6.1 Hardware Requirements

Server: Dual Core processor or higher, 4 GB RAM or higher, 50 GB storage or higher.
Client: Smartphone, tablet, laptop or desktop with internet connection.

6.2 Software Requirements

Operating System: Windows/Linux/Android/iOS
Frontend: HTML, CSS, JavaScript/React
Backend: Node.js, Express.js
Database: MySQL/MongoDB
Browser: Chrome/Firefox/Edge/Safari

6.3 Database Requirements

Entity	Important Fields
Customer	Customer_ID, Name, Email, Phone, Password
Room	Room_ID, Room_Number, Room_Type, Capacity, Price, Availability_Status
Booking	Booking_ID, Customer_ID, Room_ID, Check_In, Check_Out, Guests, Status, Total_Amount
Payment	Payment_ID, Booking_ID, Payment_Date, Amount, Payment_Method, Payment_Status
Seasonal Pricing	Pricing_ID, Season_Name, Start_Date, End_Date, Room_Type, Price

7. System Workflow

Customer: Register/Login → Search Hotel/Room → Check Availability → Select Room → Calculate Price → Enter Booking Details → Make Payment → Booking Confirmation → Generate Invoice

Admin: Admin Login → Dashboard → Manage Rooms → Manage Availability → Manage Seasonal Pricing → Manage Bookings → Manage Customers → View Payments & Reports

8. Use Case Summary

Actor	Use Cases
Customer	Register, Login, Search Rooms, Check Availability, Book Room, Make Payment, View Booking, Cancel Booking, V
Admin	Login, Manage Rooms, Manage Pricing, Manage Bookings, Manage Customers, Generate Reports

9. Constraints, Assumptions and Dependencies

Constraints

- Internet connection is required for online booking.
- Payment depends on the availability of the payment gateway.
- Room availability must be updated accurately.
- Only authorized administrators can modify hotel information.
- The system must follow applicable data-protection and payment-security requirements.

Assumptions

- Customers have valid contact information.
- Hotel room information is maintained by the administrator.
- Customers provide correct check-in and check-out dates.
- The payment service is available when payment is attempted.

Dependencies

- Internet connection
- Database server
- Web/application server
- Payment gateway
- Email/SMS service if notifications are implemented

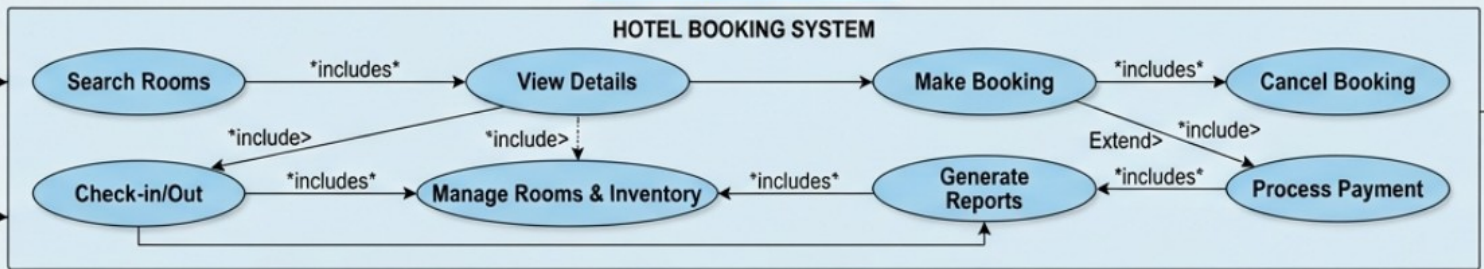
10. Expected Outcome and Conclusion

The Hotel Booking System provides an efficient way to manage hotel room reservations, availability, seasonal pricing and customer billing. It reduces manual booking work, helps avoid double booking, and provides customers with a convenient online booking experience. The SRS defines the functional and non-functional requirements needed to develop the system in a structured and reliable manner.

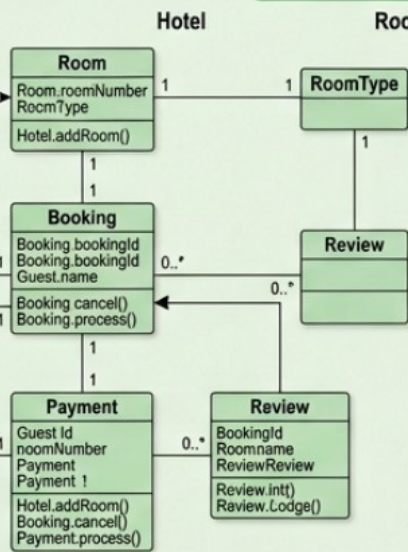
UML DIAGRAMS FOR HOTEL BOOKING SYSTEM

UML DIAGRAMS FOR HOTEL BOOKING SYSTEM

1. USE CASE DIAGRAM

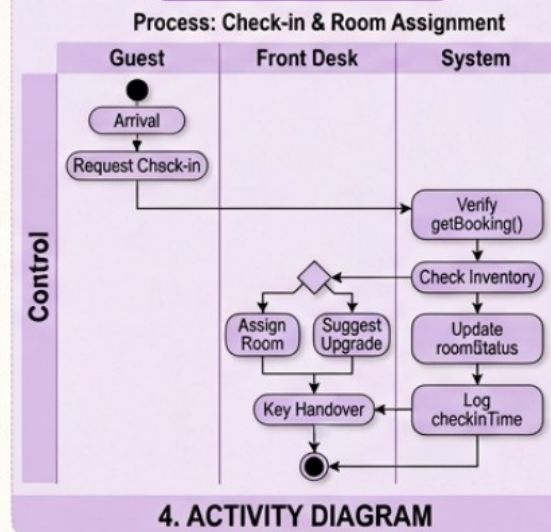


2. CLASS DIAGRAM



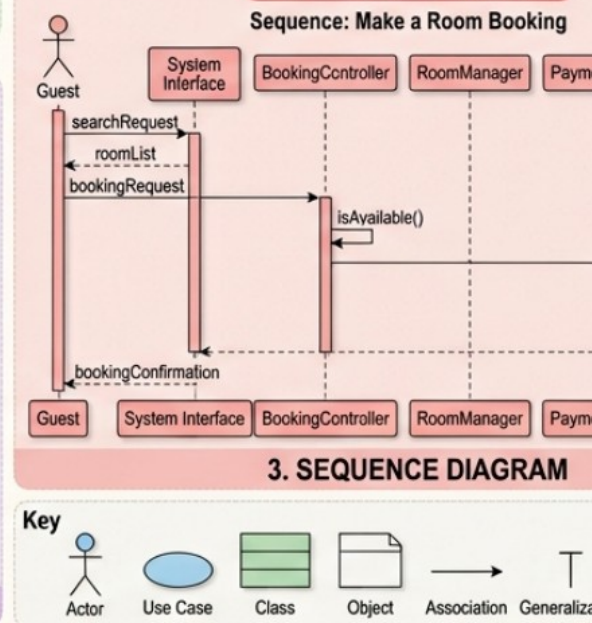
CLASS DIAGRAM

4. ACTIVITY DIAGRAM

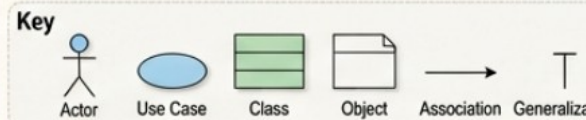


4. ACTIVITY DIAGRAM

3. SEQUENCE DIAGRAM



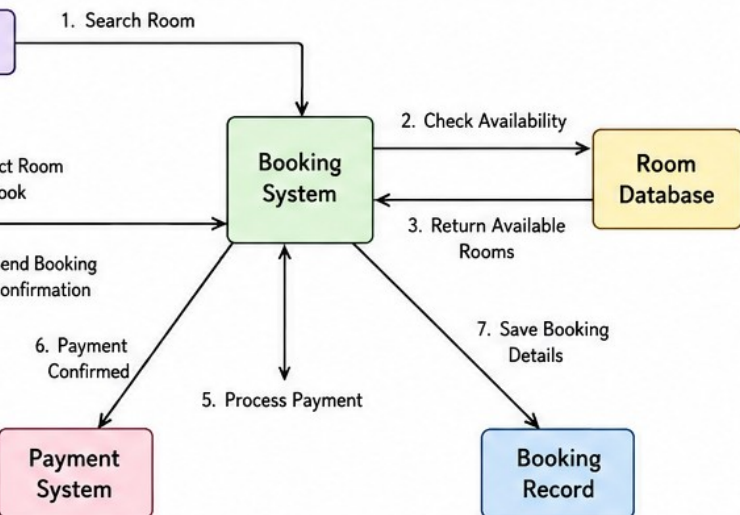
3. SEQUENCE DIAGRAM



COLLABORATION AND STATE CHART DIAGRAMS

1. COLLABORATION DIAGRAM – HOTEL BOOKING SYSTEM

Collaboration Diagram shows how objects collaborate and messages to complete a task.



SEQUENCE

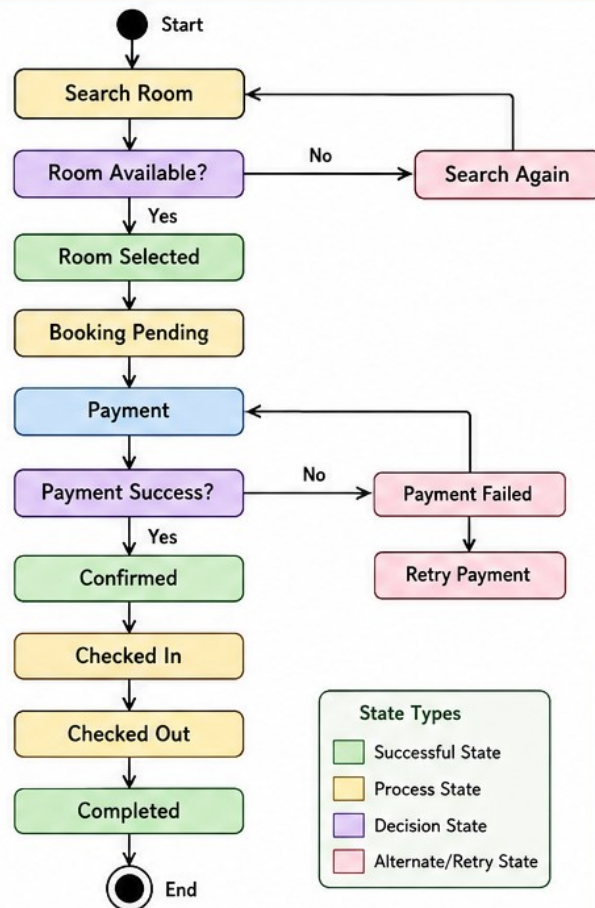
Booking System : Search room
Customer → Room Database : Check availability
Room Database → Booking System : Return available rooms
Booking System : Select room and book
Booking System → Payment System : Process payment
Payment System → Booking System : Payment confirmation
Booking System → Booking Record : Save booking
Booking System → Customer : Send booking confirmation

Objects

- Customer
- Booking System
- Room Database
- Payment System
- Booking Record

2. STATE CHART DIAGRAM – HOTEL BOOKING

State Chart Diagram shows the different states of a booking and transitions from one state to another.



State Types

- Successful State
- Process State
- Decision State
- Alternate/Retry State

Notes:

- If room is not available, customer can search again.
- If payment fails, customer can retry payment.
- After check-out, booking is completed.